



PHYS121 Integrated Science-Physics

W4T2 Rotational Kinetic Energy and Dynamics (part 1)

References:

- [1] David Halliday, Jearl Walker, Resnick Jearl, 'Fundamentals of Physics', (Wiley, 2018)
- [2] Doug Giancoli, 'Physics for Scientists and Engineers with modern physics', (Pearson, 2009)
- [3] Hugh D. Young, Roger A. Freedman, 'University Physics with Modern Physics', (Pearson, 2012)

And others specified when needed.



10.8.5. Two solid disks, which are free to rotate independently about the same axis that passes through their centers and perpendicular to their faces, are initially at rest. The two disks have the same mass, but one of has a radius R and the other has a radius $2R$. A force of magnitude F is applied to the edge of the larger radius disk and it begins rotating. What force must be applied to the edge of the smaller disk so that the angular acceleration is the same as that for the larger disk? Express your answer in terms of the force F applied to the larger disk.

a) $0.25F$

b) $0.50F$

c) F

d) $1.5F$

e) $2F$

The moment of inertia of a disk around an axis through its center of mass that is perpendicular to its surface is $\frac{MR^2}{2}$.



Learning Outcomes

- Find the work done by torques and change of kinetic energy (rotational/translational).
- Solve problems involving rolling smoothly (no slip) using Newton's second law.
- Explain the vector nature of torque and find torque using cross product.

Work and Rotational Kinetic Energy

- The rotational work-kinetic energy theorem states:

$$\Delta K = K_f - K_i = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2 = W \quad \text{Equation (10-52)}$$

- The work done in a rotation about a fixed axis can be calculated by:

$$W = \int_{\theta_i}^{\theta_f} \tau d\theta \quad \text{Equation (10-53)}$$

- Which, for a constant torque, reduces to:

$$W = \tau (\theta_f - \theta_i) \quad \text{Equation (10-54)}$$

- We can relate work to power with the equation:

$$P = \frac{dW}{dt} = \tau \omega \quad \text{Equation (10-55)}$$

Table 10-3 Some Corresponding Relations for Translational and Rotational Motion

Pure Translation (**Fixed Direction**)

Pure Rotation (**Fixed Axis**)

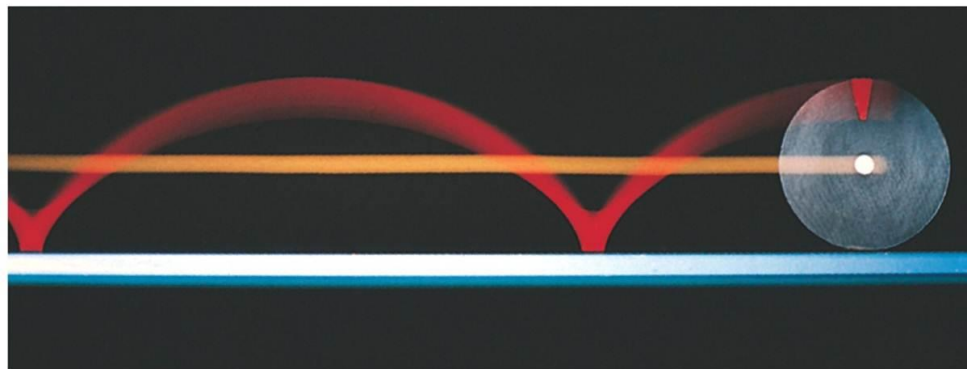
Position	x	Angular position	θ
Velocity	$v = \frac{dx}{dt}$	Angular velocity	$\omega = \frac{d\theta}{dt}$
Acceleration	$a = \frac{dv}{dt}$	Angular acceleration	$\alpha = \frac{d\omega}{dt}$
Mass	m	Rotational inertia	I
Newton's second law	$F_{\text{net}} = ma$	Newton's second law	$\tau_{\text{net}} = I\alpha$
Work	$W = \int F \, dx$	Work	$W = \int \tau \, d\theta$
Kinetic energy	$K = \frac{1}{2}mv^2$	Kinetic energy	$K = \frac{1}{2}I\omega^2$
Power (constant force)	$P = Fv$	Power (constant torque)	$P = \tau\omega$
Work–kinetic energy theorem	$W = \Delta K$	Work–kinetic energy theorem	$W = \Delta K$

Rolling as Translation and Rotation Combined

- We consider only objects that roll smoothly (no slip)
- The center of mass (CoM) of the object moves in a straight line parallel to the surface
- The object rotates around the com as it moves
- The rotational motion is defined by:

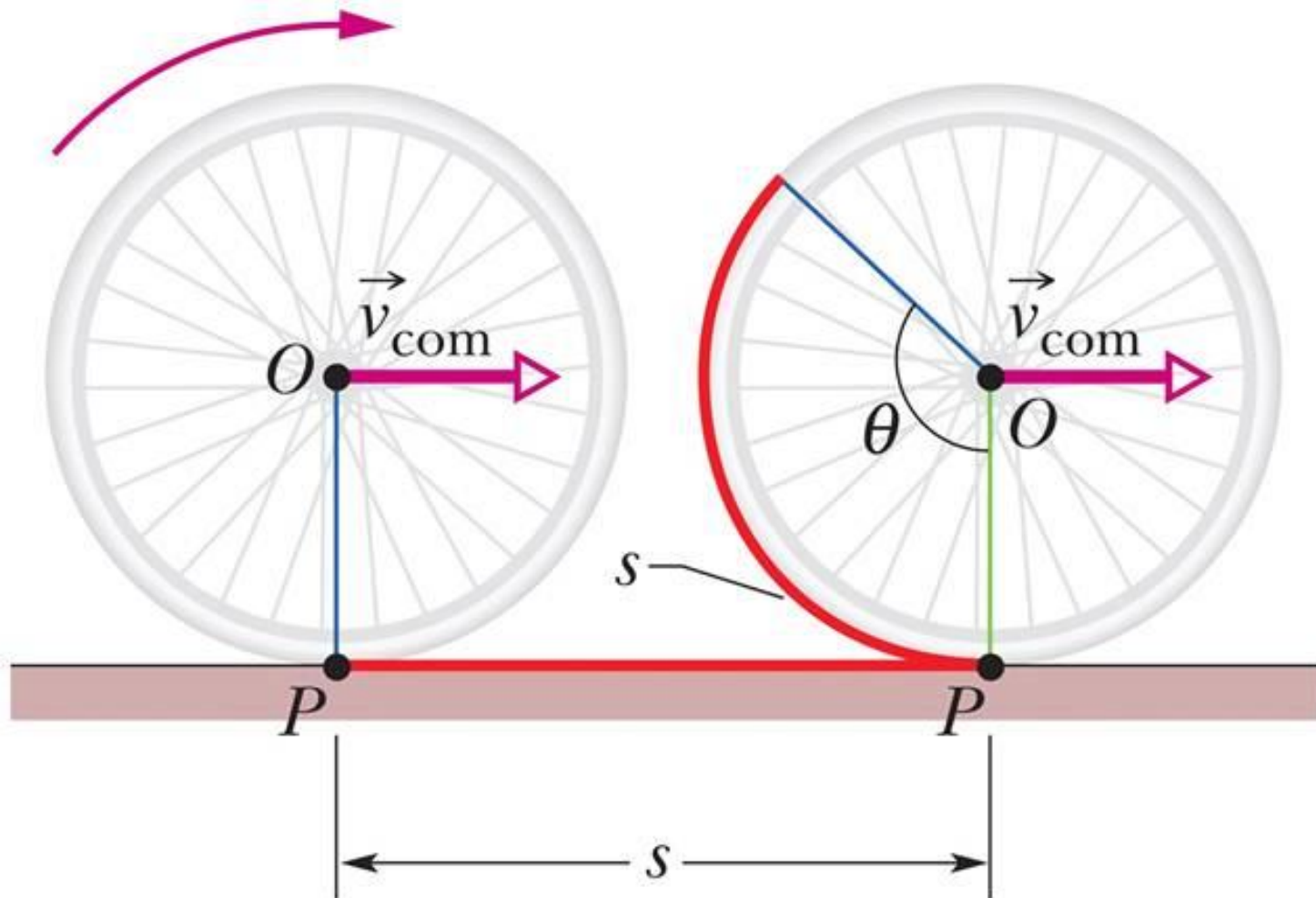
$$S = \theta R,$$

$$v_{\text{com}} = \omega R$$



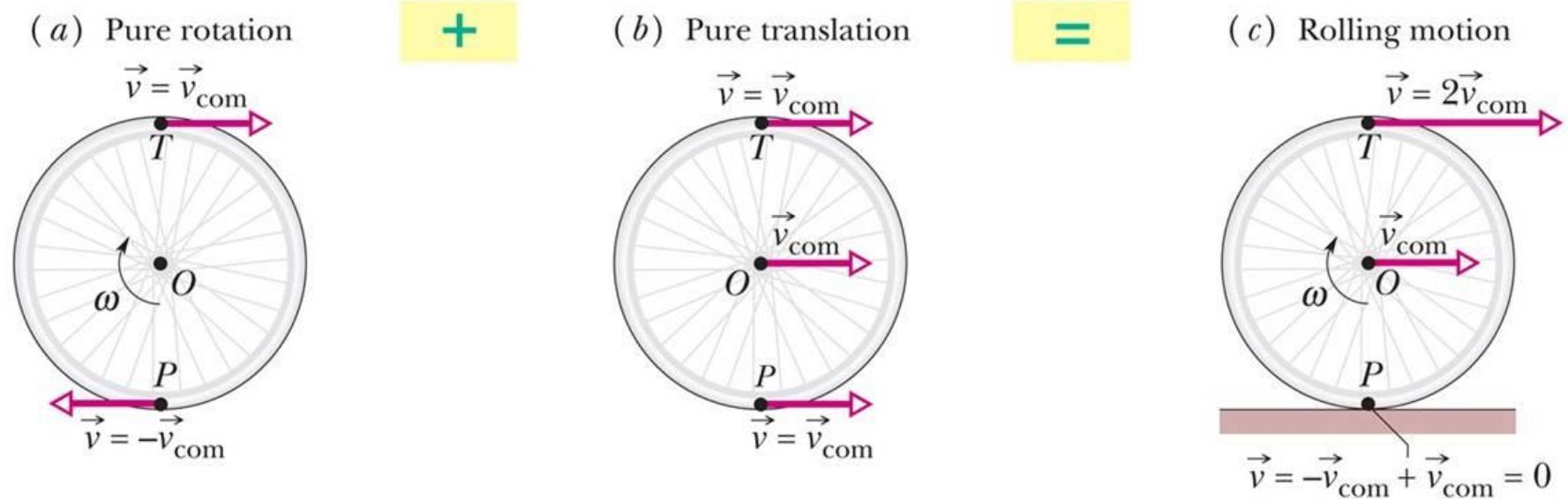
Richard Megna/Fundamental Photographs

Figure 11-2



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

Figure 11-3



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

Figure 11-4

- The figure shows how the velocities (with respect to the ground) of translation and rotation combine at different points on the wheel

Checkpoint 1

The rear wheel on a clown's bicycle has twice the radius of the front wheel, (a) When the bicycle is moving, is the linear speed at the very top of the rear wheel greater than, less than, or the same as that of the very top of the front wheel? (b) Is the angular speed of the rear wheel greater than, less than, or the same as that of the front wheel?

Answer:

(a) the same

(b) less than

Forces and Kinetic Energy of Rolling

- Combine translational and rotational kinetic energy:

$$K = \frac{1}{2} I_{\text{com}} \omega^2 + \frac{1}{2} M v_{\text{com}}^2. \quad \text{Equation (11-5)}$$

A rolling object has two types of kinetic energy: a rotational kinetic energy $\left(\frac{1}{2} I_{\text{com}} \omega^2\right)$ due to its rotation about its center of mass and a translational kinetic energy $\left(\frac{1}{2} M v_{\text{com}}^2\right)$ due to translation of its center of mass.

- If a wheel accelerates, its angular speed changes
- A force must act to prevent slip

$$a_{\text{com}} = \alpha R$$

Equation (11-6)

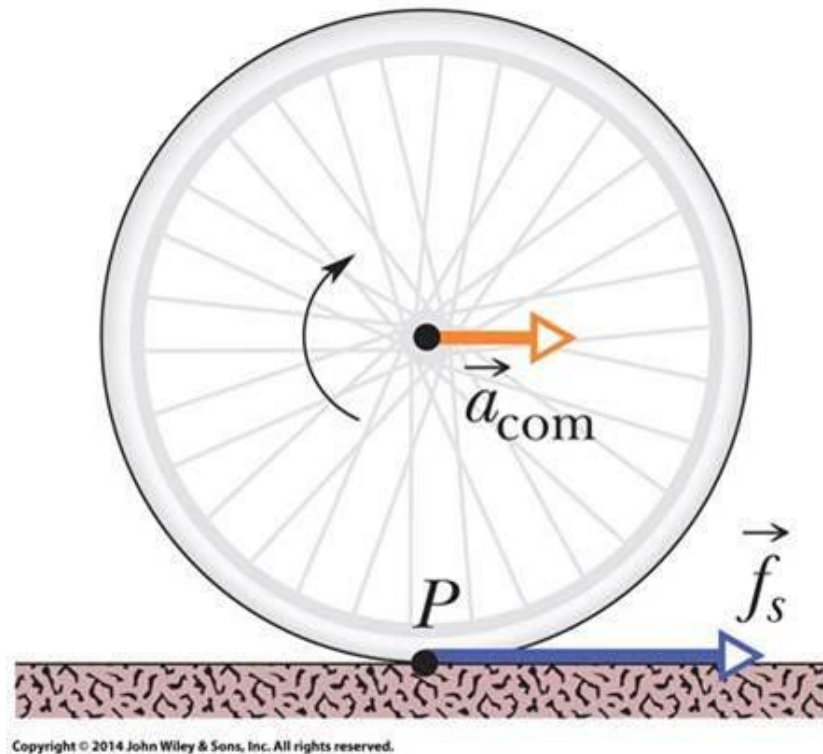
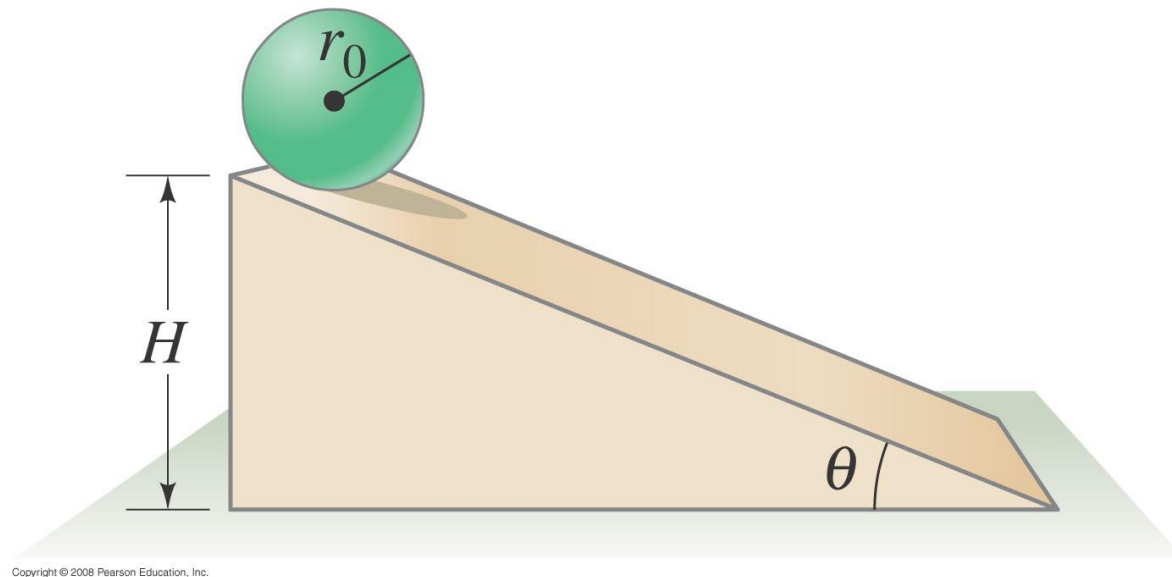


Figure 11-7 A wheel rolls horizontally without sliding while **accelerating with linear acceleration**, as on a bicycle at the start of a race. A static frictional force acts on the wheel at P , opposing its tendency to slide.

Example: Sphere rolling down an incline.

What will be the speed of a solid sphere of mass M and radius r_0 when it reaches the bottom of an incline if it starts from rest at a vertical height H and rolls without slipping?

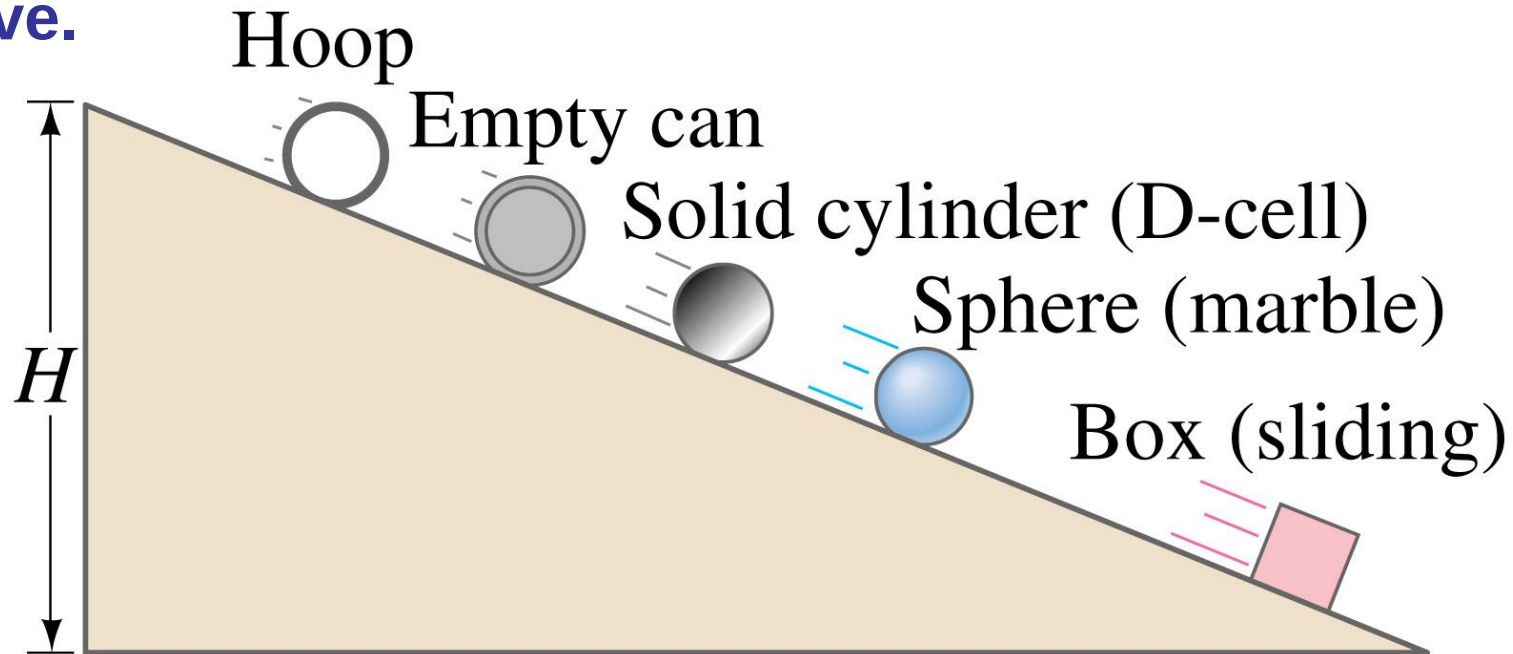
(Assume no slipping occurs because of static friction, which does no work. Think about the energy)



Copyright © 2008 Pearson Education, Inc.

When using conservation of energy, both rotational and translational kinetic energy must be taken into account.

All these objects have the same potential energy at the top, but the time it takes them to get down the incline depends on how much rotational inertia they have.



Example: Sphere rolling down an incline.

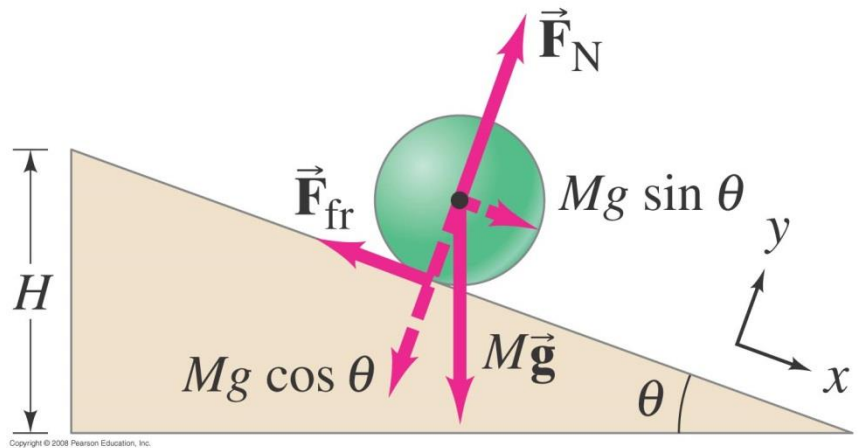
Analyze the rolling sphere of Example 10-16 in terms of forces and torques. In particular, find the velocity and the magnitude of the friction force.

$$\sum \tau = I\alpha \quad (10-14)$$

[axis fixed in inertial reference frame]

$$(\sum \tau)_{CM} = I_{CM}\alpha_{CM} \quad (10-15)$$

[axis fixed in direction, but may accelerate]



- If slip occurs, then the motion is not smooth rolling!
- For smooth rolling down a ramp:
 1. The gravitational force is vertically down
 2. The normal force is perpendicular to the ramp
 3. The force of friction points up the slope

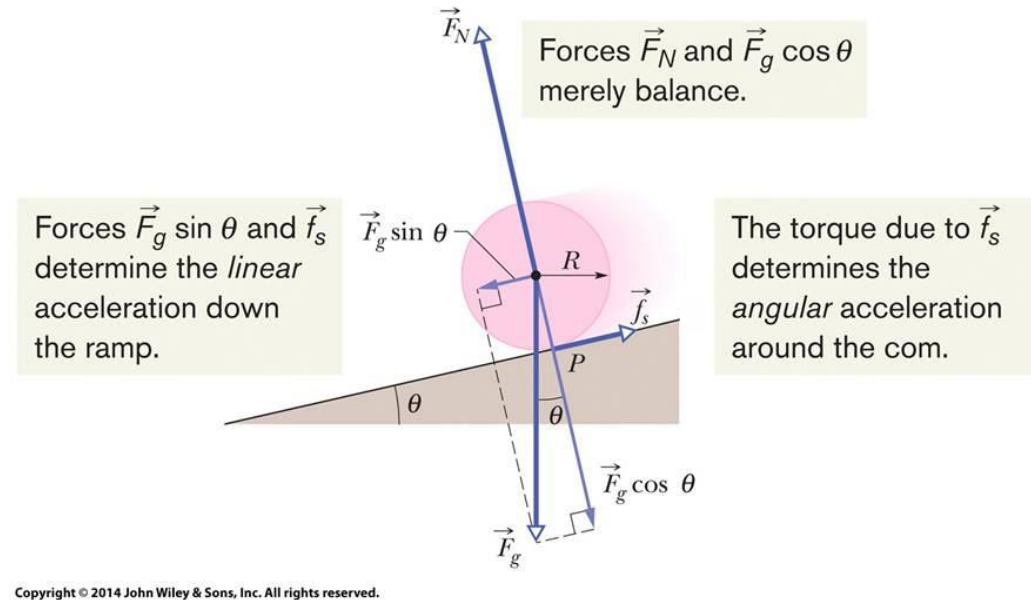


Figure 11-8

$$-f_s + Mg \sin \theta = Ma_{\text{com},x}. \quad (11.2.6)$$

$$Rf_s = I_{\text{com}}\alpha. \quad (11.2.7)$$

- We can use this equation to find the acceleration of such a body (with positive x-axis down to the ramp)

$$a_{\text{com},x} = + \frac{g \sin \theta}{1 + \frac{I_{\text{com}}}{MR^2}}. \quad \text{Equation (11-10)}$$

- Note that the frictional force produces the rotation
- Without friction, the object will simply slide

Checkpoint 1

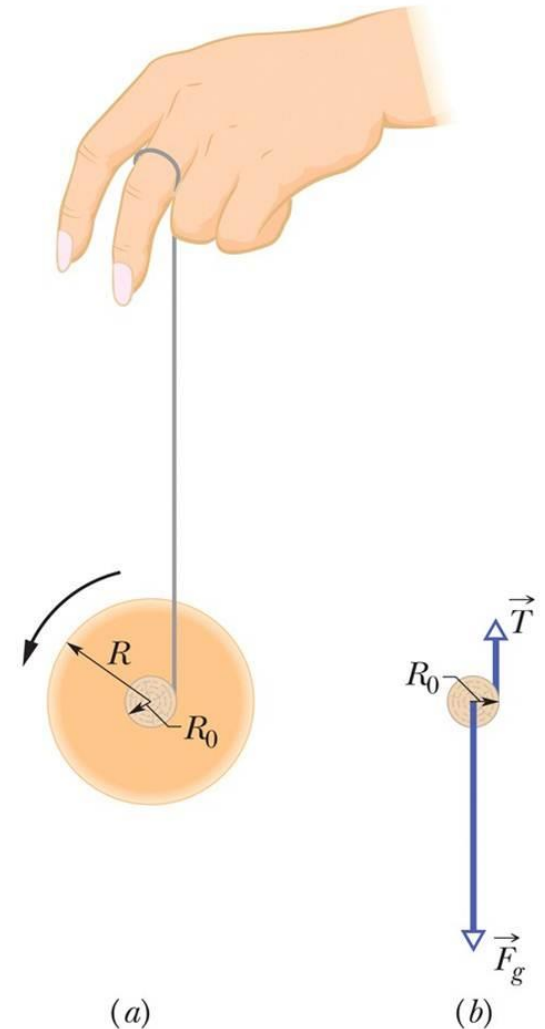
Disks A and B are identical and roll across a floor with equal speeds (angular+linear). Then disk A rolls up an incline, reaching a maximum height h , and disk B moves up an incline that is identical except that it is frictionless. Is the maximum height reached by disk B greater than, less than, or equal to h ?

Answer:

The maximum height reached by B is less than that reached by A . For A , all the kinetic energy becomes potential energy at h . Since the ramp is frictionless for B , all of the rotational K stays rotational, and only the translational kinetic energy becomes potential energy at its maximum height.

The Yo-Yo

- As a yo-yo moves down a string, it loses potential energy mgh but gains rotational and translational kinetic energy
- To find the linear acceleration of a yo-yo accelerating down its string:
 1. Rolls down a “ramp” of angle 90°
 2. Rolls on an axle instead of its outer surface
 3. Slowed by tension T rather than friction



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved

Figure 11-9

- Replacing the values in 11-10 leads us to (positive up vertically):

$$a_{\text{com}} = -\frac{g}{1 + \frac{I_{\text{com}}}{MR_0^2}}, \quad \text{Equation (11-13)}$$

Example Calculate the acceleration of the yo-yo

- $M = 150 \text{ grams}, R_0 = 3 \text{ mm}, I_{\text{com}} = \frac{Mr^2}{2} = 3\text{E} - 5 \text{ kg m}^2$

- Therefore
$$a_{\text{com}} = \frac{-9.8 \text{ m/s}^2}{\left(\frac{1 + 3\text{E} - 5}{(0.15 * 0.003^2)} \right)}$$
$$= -.4 \text{ m/s}^2$$

What about the tension in the cord?

Torque Revisited

- Previously, torque was defined only for a rotating body and a fixed axis
- Now we redefine it for an individual particle that moves along any path relative to a fixed point
- The path need not be a circle; torque is now a vector
- Direction determined with *right-hand-rule*

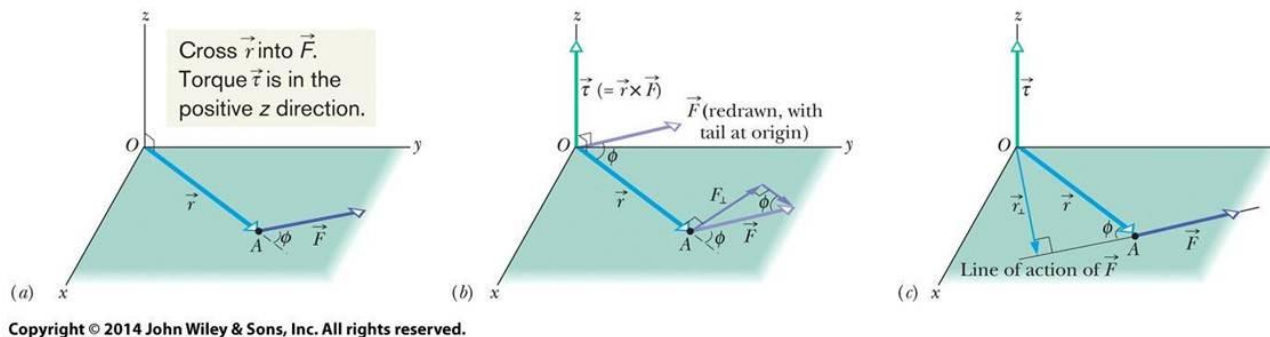


Figure 11-10

- The general equation for torque is:

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \text{Equation (11-14)}$$

- We can also write the magnitude as:

$$\tau = rF \sin \phi, \quad \text{Equation (11-15)}$$

- Or, using the perpendicular component of force or the moment arm of F :

$$\tau = rF_{\perp}, \quad \text{Equation (11-16)}$$

$$\tau = r_{\perp} F, \quad \text{Equation (11-17)}$$

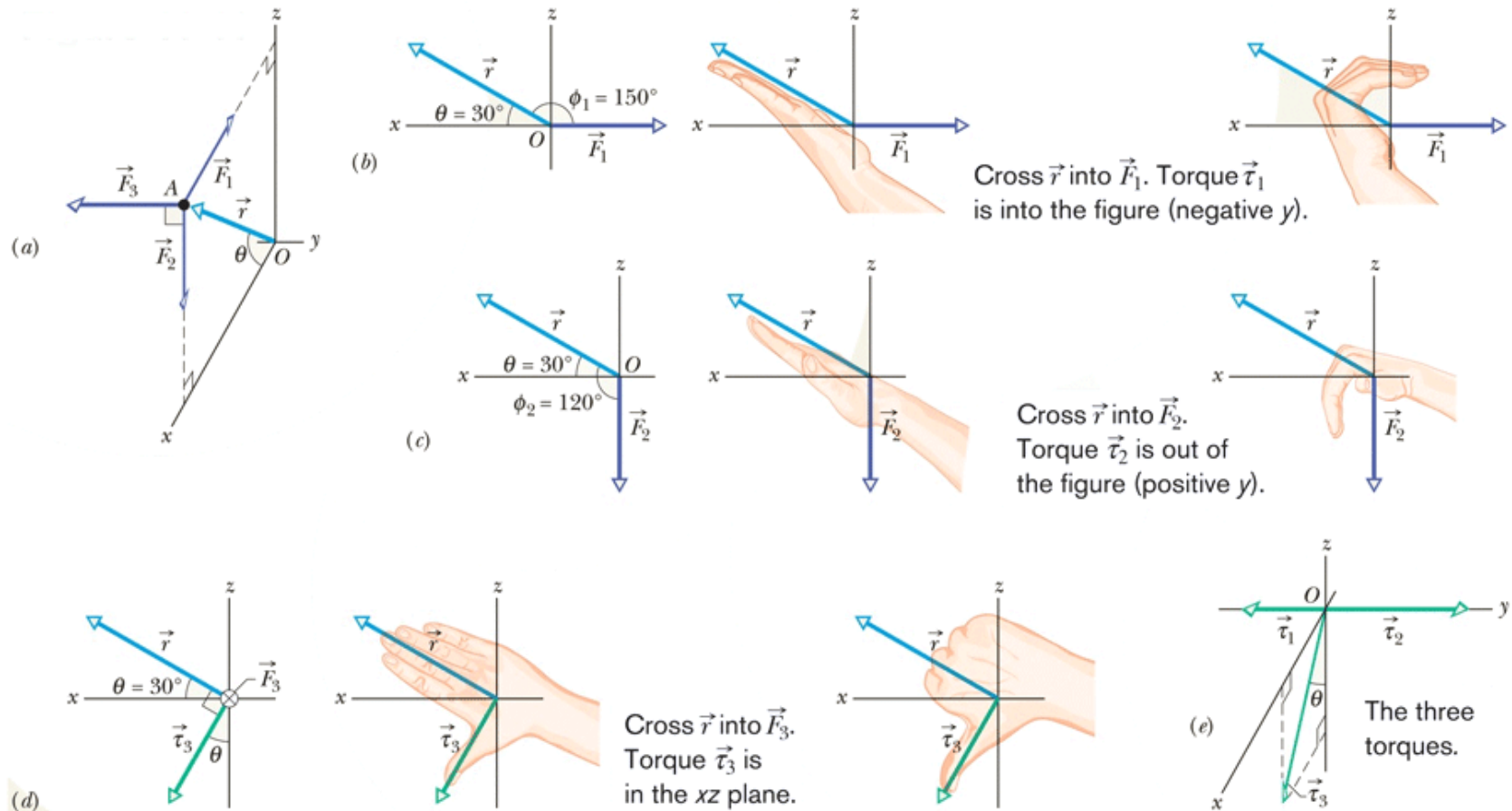
Checkpoint 3

The position vector $\vec{r}$ of a particle points along the positive direction of a z axis. If the torque on the particle is (a) zero, (b) in; the negative direction of x , and (c) in the negative direction of y , in what direction is the force causing the torque?

Answer:

- (a) along the z direction
- (b) along the $+y$ direction
- (c) along the $+x$ direction

Example Calculating net torque:



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

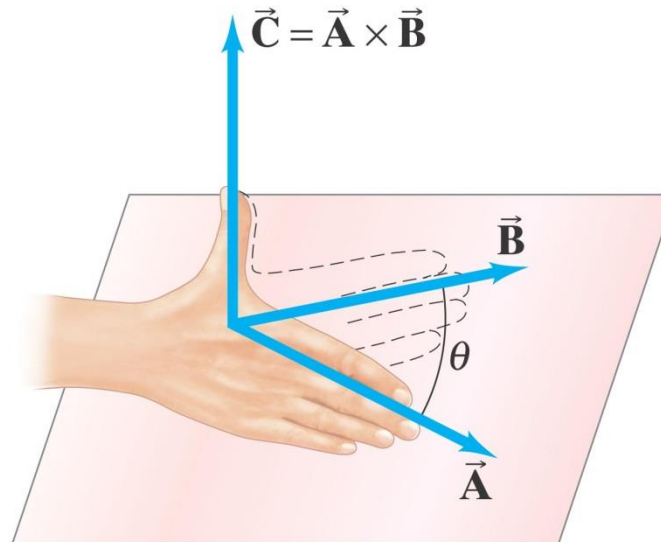
Figure 11-11

Vector Cross Product (revisited); Torque as a Vector

The vector cross product is defined as:

$$C = |\vec{\mathbf{A}} \times \vec{\mathbf{B}}| = AB \sin \theta.$$

The direction of the cross product is defined by a **right-hand rule**:



The cross product can also be written in determinant form (in **components**):

$$\begin{aligned}
 \vec{\mathbf{A}} \times \vec{\mathbf{B}} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} \\ A_x & A_y \\ B_x & B_y \end{vmatrix} \\
 &= (A_y B_z - A_z B_y) \hat{\mathbf{i}} + (A_z B_x - A_x B_z) \hat{\mathbf{j}} + (A_x B_y - A_y B_x) \hat{\mathbf{k}}.
 \end{aligned}$$

Example: Torque vector.

Suppose the vector $\vec{r}$ is in the xz plane, and is given by $\vec{r} = (1.2 \text{ m})\hat{i} + (1.2 \text{ m})\hat{k}$. Calculate the torque vector $\vec{\tau}$ if $\vec{F} = (150 \text{ N})\hat{i}$

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1.2 & 0 & 1.2 \\ 150 & 0 & 0 \end{pmatrix} = 180 \text{ N} \cdot \text{m} \hat{j}$$

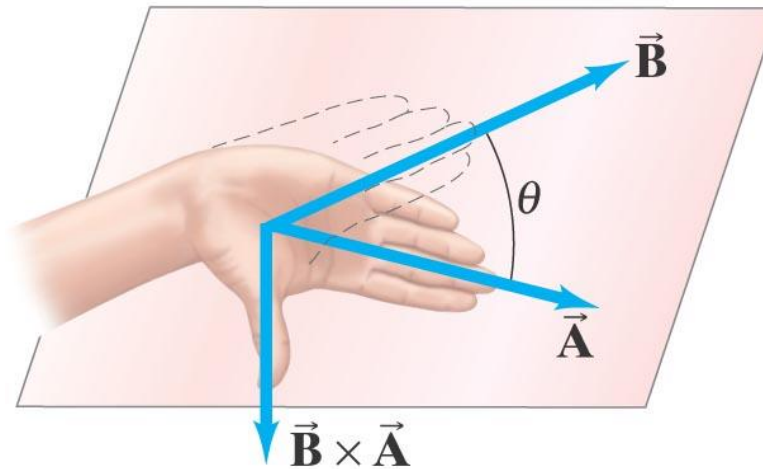
Some properties of the cross product:

$$\vec{\mathbf{A}} \times \vec{\mathbf{A}} = 0$$

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = -\vec{\mathbf{B}} \times \vec{\mathbf{A}}$$

$$\vec{\mathbf{A}} \times (\vec{\mathbf{B}} + \vec{\mathbf{C}}) = (\vec{\mathbf{A}} \times \vec{\mathbf{B}}) + (\vec{\mathbf{A}} \times \vec{\mathbf{C}})$$

$$\frac{d}{dt}(\vec{\mathbf{A}} \times \vec{\mathbf{B}}) = \frac{d\vec{\mathbf{A}}}{dt} \times \vec{\mathbf{B}} + \vec{\mathbf{A}} \times \frac{d\vec{\mathbf{B}}}{dt}.$$



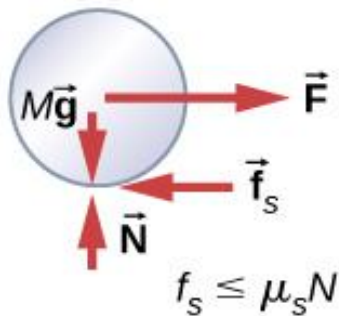


Learning Outcomes

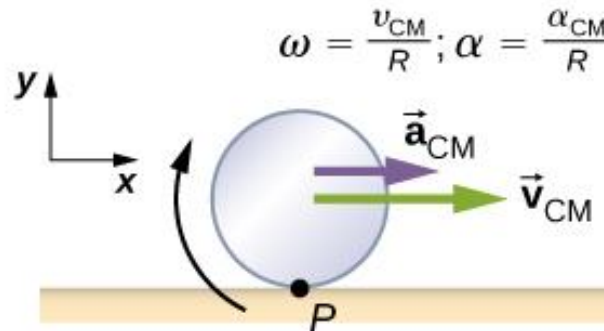
- ✓ Find the work done by torques and change of kinetic energy (rotational/translational).
- ✓ Solve problems involving rolling smoothly (no slip) using Newton's second law.
- ✓ Explain the vector nature of torque and find torque using cross product.

CHAPTER REVIEW AND EXAMPLES, DEMOS

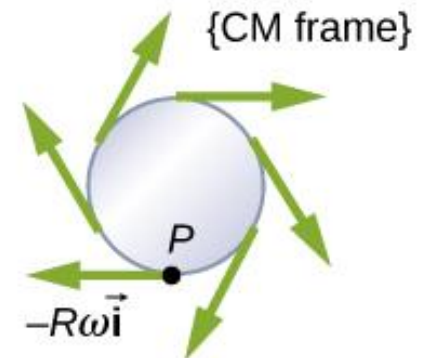
<https://openstax.org/books/university-physics-volume-1/pages/11-summary>



(a) Forces on the wheel



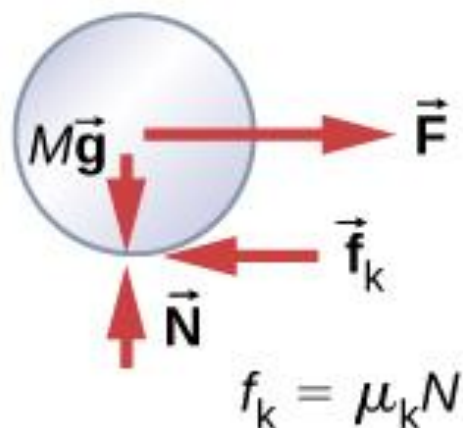
(b) Wheel rolls without slipping



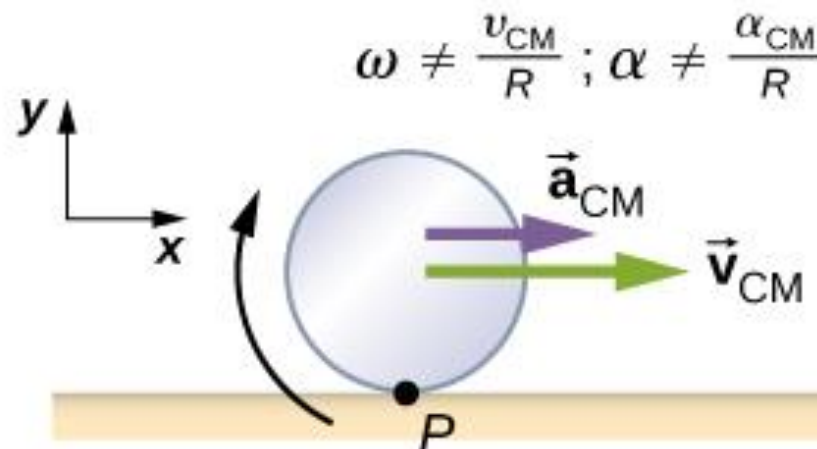
(c) Point P has velocity vector in the negative direction with respect to the center of mass of the wheel

- (a) A wheel is pulled across a horizontal surface by a force $\vec{F}$. The force of static friction $\vec{f}_s$, $|\vec{f}_s| \leq \mu_s N$ is large enough to keep it from slipping.
- (b) The linear velocity and acceleration vectors of the center of mass and the relevant expressions for ω and α . Point P is at rest relative to the surface.
- (c) Relative to the center of mass (CM) frame, point P has linear velocity $-R\omega\hat{i}$.

FIGURE 11.6



(a) Forces on wheel



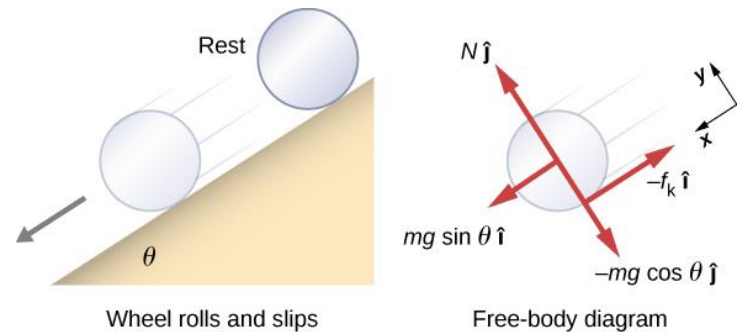
(b) Wheel is rolling and slipping

- (a) Kinetic friction arises between the wheel and the surface because the wheel is slipping.
- (b) The simple relationships between the linear and angular variables are no longer valid.

Example 11.2

Rolling Down an Inclined Plane with Slipping

A solid cylinder rolls down an inclined plane from rest and undergoes slipping (**Figure 11.7**). It has mass m and radius r . (a) What is its linear acceleration? (b) What is its angular acceleration about an axis through the center of mass?



Questions

